

Scripts for clustering multi-variate datasets and analysing results in MATLAB using agglomerative hierarchical clustering.

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Reference:

Ireland, B., Biggs, J., and Anantrasirichai, N. (2026) *Clustering global volcano deformation datasets: insights and limitations for analogue signals*. Submitted

Quick start:

Note: Before running *RecreateClusteringResults.m* or *Cluster_Own_Data.m*, ensure the current working directory is “*/Scripts_and_Results” (use pwd command to check, and cd command to change if needed).

1. I want to reproduce the figures and clustering from the manuscript

- a. In MATLAB, open the file “Scripts_and_Results → RecreateClusteringResults.m”
- b. Run RecreateClusteringResults.m script. Plots will appear as in the paper, and results can be found in the Cluster_Validation_Results, ClusterStats, Paper_Clustering_Results, and Re_Clustered_Results folders. See **Files and file structure** section for more detail.

2. I want to try different agglomerative hierarchical clustering approaches on the same dataset

- a. In MATLAB, open the file “Scripts_and_Results → Cluster_Own_Data.m”
- b. Ensure the line “`Catalogue = 'Paper_Dataset_Example.xlsx';`” is uncommented (Line 22).
- c. On lines 86 and 87, change DistMethod and LinkMethod to your desired distance metric and linkage criteria. See <https://uk.mathworks.com/help/stats/pdist.html> and <https://uk.mathworks.com/help/stats/linkage.html> for all options
- d. Run Cluster_Own_Data.m script. Visualisation validation and clustering plots will appear and results can be found in the Example_Results folder. See **Files and file structure** section for more detail.

3. I want to see the input volcano deformation catalogue

- a. Download the “Deformation_Catalogues → Deformation_Catalogue.xlsx” file. There are alternative versions in the same folder showing only the events used in the clustering, and removing any interpolated values, see **Files and file structure** section for more detail.
- b. The volcano deformation events in these catalogues come from common events between the catalogues of Biggs and Pritchard (2017), and Ebmeier *et al.* (2018).

4. I want to apply the same clustering approaches to my own multi-variate dataset

- a. Prepare your dataset in the format shown in the file “Deformation_Catalogues → Example_Dataset → Example_Input_Dataset.xlsx”.
- b. In MATLAB, open the file “Scripts_and_Results → Cluster_Own_Data.m”
- c. On Line 21, specify your input dataset .xlsx file prepared in the previous step.
- d. If you do not have any background information and/or ground truth, uncomment lines 29 and 30.

- e. Specify your clustering options, run name, and output folder in lines 34-74 following descriptions in the file.
- f. Run `Cluster_Own_Data.m` script. Visualisation validation and clustering plots will appear and results can be found in the `Example_Results` folder. See **Files and file structure** section for more detail.

5. I want to do something else

- a. Please see **Files and file structure** and individual MATLAB scripts and functions for more detail. If you have a specific request, you can contact the author at the address at the top of the this document.

Files and file structure:

Key: Folders | files

- **Scripts_and_Results**
 - `Cluster_Own_Data.m` – MATLAB script to run to cluster your own dataset in the format shown in datasets in the **Example_Dataset** folder.
 - `RecreateClusteringResults.m` – MATLAB script to recreate the figures from the original manuscript from the datasets in the **Deformation_Catalogues** folder.
 - **Cluster_Validation_Results**
 - `ZX_Data_*.mat` – Cluster validation results for tested combinations of distance metric and linkage criteria Linkage (Z) and input dataset (X). For X, columns are different parameters, rows are different events.
 - **Clustering** - MATLAB functions for clustering validation and analysis, used by `Cluster_Own_Data.m` and `RecreateClusteringResults.m`
 - **ClusterStats**
 - `ClusterStats_*.xlsx` – Cluster statistics from the manuscript for each cluster, each sheet represents a different parameter.
 - **Colormaps** – Third-party colormaps used during visualisation of clustering results
 - **Example_Results**
 - **Approach_Validation**
 - `ZX_Data_*.mat` - Cluster validation results for tested combinations of distance metric and linkage criteria Linkage (Z) and example input dataset (X). For X, columns are different parameters, rows are different events.
 - **Clustering**
 - `ZX_Data_*.mat` - Cluster validation results for the chosen combination of distance metric and linkage criteria Linkage (Z) and example input dataset (X). For X, columns are different parameters, rows are different events.
 - `Clusters_*.xlsx` – Row of cluster assignments for a given clustering run and number of clusters, specified in the filename. The rows in this list correspond to rows in the input dataset.
 - `Stats_*.xlsx` – Cluster statistics from a given clustering run and number of clusters, specified in the filename. Statistics given for each cluster, each sheet represents a different parameter.
 - **Paper_Clustering_Results**
 - `ZX_Data_*.mat` – Clustering results presented in the manuscript with Linkage (Z) and input dataset (X). For X, columns are different parameters, rows are different events.
 - **Plotting** - MATLAB functions for visualising clustering results
 - **Re_Clustered_Results**
 - `ZX_Data_*.mat` – Clustering results presented in the manuscript with Linkage (Z) and input dataset (X), when re-clustering the larger Clusters 1 and 4. For X, columns are different parameters, rows are different events.
- **GVP**

- *GVP_Volcano_List_Holocene.xlsx* – Global Volcanism Programme (GVP) list of Holocene active volcanoes, used as background information.
- *GVP_Volcano_List_Pleistocene.xlsx* – Global Volcanism Programme (GVP) list of Pleistocene active volcanoes, used as background information.
- **Deformation Catalogues**
 - *Deformation_Catalogue.xlsx* – Deformation catalogue of events created for the manuscript and used in the clustering. Used by the *RecreateClusteringResults.m* script. See manuscript for more details.
 - *Deformation_Catalogue_No_Interpolation.xlsx* – Deformation catalogue of events created for the manuscript and used in the clustering, with any interpolated values removed. Used by the *RecreateClusteringResults.m* script. See manuscript for more details.
 - *Clustered_Events_Only.xlsx* – Same as *Deformation_Catalogue.xlsx* but cropped to only include the 137 events that had reported values for all characteristics used in the clustering.
 - **Example Dataset**
 - *Example_Input_Dataset.xlsx* – Example toy dataset to demonstrate the input format for the *Cluster_Own_Data.m* script and the related output plots and files.
 - *Paper_Dataset_Example.xlsx* – The clustered dataset from the manuscript and in the *Clustered_Events_Only.xlsx* and *Deformation_Catalogue.xlsx* files, converted into the input format to work with the *Cluster_Own_Data.m* script.

References:

Biggs, J., & Pritchard, M. E. (2017). Global volcano monitoring: what does it mean when volcanoes deform?. *Elements*, 13(1), 17-22.

Ebmeier, S. K., Andrews, B. J., Araya, M. C., Arnold, D. W. D., Biggs, J., Cooper, C., ... & Williamson, J. L. (2018). Synthesis of global satellite observations of magmatic and volcanic deformation: implications for volcano monitoring & the lateral extent of magmatic domains. *Journal of Applied Volcanology*, 7(1), 2.

For more information:

Clustering in MATLAB - <https://uk.mathworks.com/help/stats/hierarchical-clustering.html>

Global Volcanism Programme - <https://volcano.si.edu/>